

ReproBench: Benchmarking LLM Agents on CVE-only Vulnerability Reproduction

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Abstract

Large language model (LLM) agents are increasingly evaluated on cybersecurity tasks, yet most benchmarks begin after the target environment has already been prepared: the agent receives source code, a patch, a running service, a container, or an executable binary. This setting leaves open a central question for real-world vulnerability analysis: can an agent reconstruct the missing environment needed to reproduce a known vulnerability from scratch? We introduce REPROBENCH, a public-evidence-grounded benchmark for from-scratch reproduction attempts of IoT firmware vulnerabilities from only a CVE identifier. REPROBENCH contains 30 manually curated IoT CVEs, balanced across buffer overflows, command injections, and authentication bypasses. Each task is accompanied by a manually inspected public-evidence dossier built from CVE records, NVD entries, vendor advisories, public PoCs, exploit databases, blogs, and firmware references. Agents must traverse a six-stage vulnerability reproduction pipeline: information gathering, firmware acquisition, firmware extraction, binary identification, service rehosting, and vulnerability triggering. We propose an evidence-grounded, non-binary scoring framework that separately evaluates planning and execution and requires observable artifacts for every non-zero score. Our evaluation consists of 450 reproduction runs (30 CVEs, 5 models, 3 trials) scored under a best-of-three protocol. Results show that agents average 50.1/100 overall, but only 4.7% of CVE-model pairs achieve real-target vulnerability triggering; 69.8% of runs substitute simulated reproductions for real firmware execution. REPROBENCH exposes the gap between textual vulnerability understanding and grounded firmware execution, revealing that the rehosting stage is the primary bottleneck where agents lose real-target grounding.

Introduction

LLM agents now plan, call tools, write code, and operate in software repositories with increasing autonomy, producing benchmarks for software engineering and cybersecurity (Jimenez et al. 2024; Yang et al. 2026; Wang et al. 2025; Zhu et al. 2025; Lee et al. 2026; Zhang et al. 2026; Wang et al. 2026; Lee and Brumley 2026; Pu et al. 2026). A common assumption: the world has already been prepared for the

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agent. The agent receives a repository, patch, container, harness, or running service. Known-vulnerability reproduction in the wild begins in a less structured state: an analyst may start with only a CVE identifier and must discover the affected product, locate the correct artifact, recover a runnable environment, and validate vulnerability behavior against the real target. IoT firmware makes this gap especially sharp—firmware is a proprietary binary blob compiled for MIPS or ARM, packed in vendor-specific formats, dependent on missing NVRAM state, and reachable only after rehosting under emulation. Firmware reproduction is not merely exploit writing; it is an end-to-end artifact reconstruction and validation task.

We introduce REPROBENCH, a benchmark that evaluates whether LLM agents can move from a bare CVE identifier toward auditable evidence of IoT firmware vulnerability reproduction. REPROBENCH moves beyond prepared environments to evaluate the full artifact-to-execution pipeline, stressing capabilities that prepared-environment benchmarks hide: open-world information gathering, artifact provenance checking, cross-architecture binary analysis, long-horizon tool orchestration, and grounded validation. Each task is grounded in a manually inspected *public-evidence dossier*: a reference packet of public vulnerability facts and expected evidence types. Agents receive only the CVE identifier, not the dossier.

REPROBENCH also addresses a methodological problem: plausible-looking reproductions are easy to fabricate. When rehosting is difficult, agents produce substitutes such as Python mock servers or synthetic C programs that crash. REPROBENCH uses non-binary, evidence-grounded scoring across six stages with real-target gating for rehosting and triggering.

Our contributions are:

- We formulate CVE-only vulnerability reproduction as a from-scratch, long-horizon LLM agent benchmark.
- We construct a 30-CVE public-evidence dataset balanced across buffer overflow, command injection, and authentication bypass.
- We propose an evidence-grounded scoring framework with real-target gating across six dependent stages.
- We evaluate 450 runs: agents average 50.1/100 overall, but only 4.7% achieve real-target triggering; 69.8% involve

simulation substitution.

Motivation and Task

Why From-Scratch Reproduction?

Existing cybersecurity benchmarks measure post-environment capabilities. CyberGym and SEC-bench provide source-level projects (Wang et al. 2025; Lee et al. 2026); CVE-Bench provides prepared web targets (Zhu et al. 2025); ExploitBench evaluates after a target is available (Lee and Brumley 2026); K-Repro starts from kernel patch and buildable kernel (Pu et al. 2026). None answer whether agents can create the reproduction environment themselves.

REPROBENCH asks that question directly: the input is a CVE identifier only; the goal is auditable evidence that the agent progressed toward reproducing the vulnerability on the real affected firmware service. This makes the task open-world, long-horizon, and artifact-grounded.

Why IoT Firmware?

IoT firmware vulnerabilities provide a compact stress test: the agent must retrieve public information, verify version details, locate firmware that may have disappeared, distinguish real firmware from HTML errors, unpack vendor-specific formats, reason about non-x86 binaries, and use rehosting tools. Each stage leaves auditable artifacts, making the domain suitable for fine-grained evaluation.

Six-Stage Pipeline

REPROBENCH defines six dependent stages: R1 Information Gathering, R2 Firmware Acquisition, R3 Firmware Extraction, R4 Binary Identification, R5 Service Rehosting, and R6 Vulnerability Triggering. The dependency order is strict: R6 is valid only if the real service from R5 is reachable, and R5 is meaningful only after the correct firmware and binary have been identified. This structure is essential for rejecting simulation substitution.

Benchmark Construction

Manual CVE Curation

We construct REPROBENCH through manual curation from public CVE databases, vendor advisories, NVD entries, exploit databases, security blogs, and GitHub PoC repositories. A candidate is retained if it affects an IoT or embedded firmware product, public references identify the affected product or firmware family, the vulnerability class is assignable to our target categories, and there is enough public evidence to build a reference dossier. From the candidate pool, we select 30 CVEs balanced equally across three classes: 10 buffer overflows, 10 command injections, and 10 authentication bypasses. These classes require different validation signals—crashes for buffer overflows, command side effects for injection, and unauthenticated access for bypass—creating a controlled benchmark for comparing agent capabilities.

Public-Evidence Dossiers

For each CVE, we construct a public-evidence dossier containing vendor and product information, affected firmware versions, vulnerability class, affected endpoint, public references, PoC descriptions, firmware acquisition hints, target binary names, architecture, and expected trigger evidence type. The dossier is used only for scoring; agents receive only the CVE identifier. The dossier is not a claim that the authors have independently reproduced every CVE end to end. Rather, it defines the publicly documented facts and evidence requirements against which an agent’s artifacts are audited.

Evidence Scope and Attribution

The dossier is organized around the six stages. R1 fields describe checkable public facts; R2–R4 fields describe artifact expectations; R5–R6 fields specify acceptable evidence types for real execution (e.g., service banner, crash signal, command side effect) without providing an exploit oracle. This design separates three cases: agent capability gaps (fail despite strong evidence), evidence insufficiency (stale or missing public artifacts), and simulation substitution (plausible reports without dossier-supported artifacts). We use a conservative attribution rule: *artifact unavailable* requires confirmed stale sources plus a reasonable acquisition attempt; *agent acquisition failure* requires accessible sources but inadequate search; borderline cases are excluded from model-only failure counts.

Evidence-Grounded Scoring

Why Non-Binary Scoring?

Binary success is poorly suited to from-scratch firmware reproduction. Most runs fail before the final trigger, but a run that downloads and extracts the correct firmware has progressed much further than one that only reads an advisory. Conversely, an agent can fabricate a mock service and crash it, producing a convincing report without touching the real target. REPROBENCH therefore scores progress stage by stage and requires observable evidence for every non-zero score.

Plan Score

Before execution, the agent writes a reproduction plan. The plan score has three components, each normalized to 0–100: $\text{Plan} = 0.6 \times \text{Coverage} + 0.3 \times \text{Dependency} + 0.1 \times \text{Fallback}$. *Coverage* evaluates whether the plan explicitly covers all six phases with the same weights as the task score. *Dependency correctness* checks that the planned order respects $R1 \rightarrow R2 \rightarrow R3 \rightarrow R4 \rightarrow R5 \rightarrow R6$, deducting 20 points per violation. *Fallback planning* is scored on a quality scale from 0 (no fallback) to 100 (phase-specific contingencies with blocked phase, failure mode, and alternative action).

Task Score

The task score assigns 100 points across R1–R6: 15 points each for R1–R4 and 20 points each for R5–R6. Each phase is scored directly from its phase-specific evidence checklist; credit requires observable artifacts aligned with the

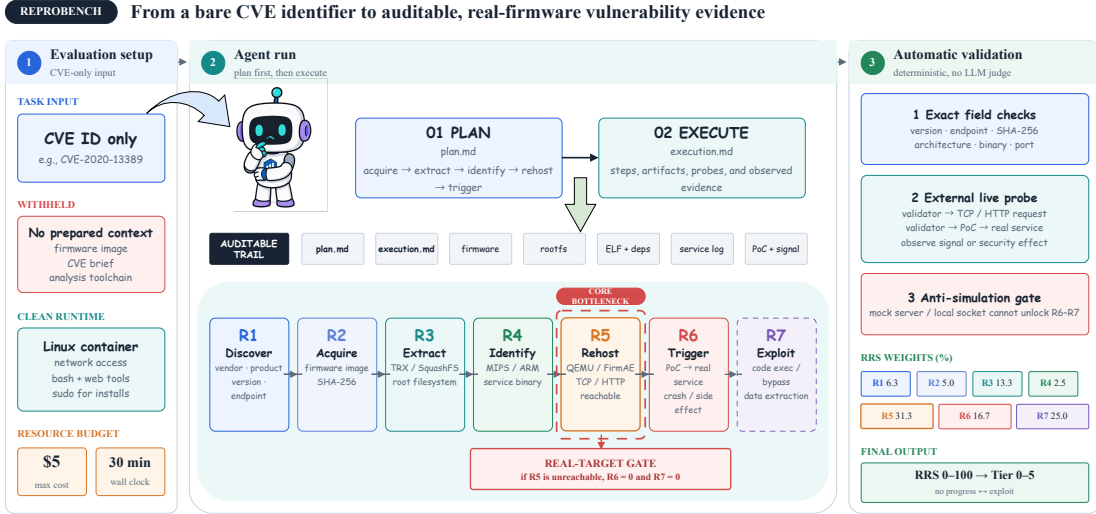


Figure 1: Scoring framework overview. Each run is scored across six phases using evidence checklists, with real-target gating at R5/R6.

public-evidence dossier. For example, R2 checks firmware acquisition, version verification (using version strings, meta-data, checksum or hash), and source address verification; R5 checks environment setup, binary launch, dependency initialization, service reachability, and real-firmware-service confirmation. Plan consistency, recovery attempts, and efficiency are recorded in failure attribution but do not receive separate task score points.

The overall score is $\text{Overall} = 0.2 \cdot \text{Plan} + 0.8 \cdot \text{Task}$. We weight execution more heavily because the benchmark measures whether an agent can realize a reproduction, not merely describe one.

Real-Target Gating

R5 and R6 enforce real-target validity: the reproduction task requires running the real vulnerable binary from the extracted firmware. Any approach that does not run the real firmware binary—mock servers, harnesses transcribing vulnerable logic, host-native simulations, or static-only analysis—is considered simulated reproduction and receives 0 for R5 and R6. This is not a penalty; the simulation work simply has no value for real-target reproduction. If the agent attempted real rehosting before switching to simulation, the real attempt is scored on its progress and the simulation is ignored.

Scoring Skill and Audit

We use a rubric-constrained scoring skill to scale evaluation to 450 runs. The scorer receives the dossier, plan, trace, workspace, and reports for a single run, cites concrete evidence for every non-zero score, and outputs structured JSON. A stratified human audit validates the scoring on a sample of runs—focusing on non-zero R5/R6 claims, high scores, and simulation-like workspaces—but does not add hidden credit for unobserved work.

Experimental Design

Agent Runs

We evaluate five model configurations over the 30-CVE benchmark with three independent trials per CVE, for 450 reproduction runs: `claude-sonnet-4-6`, `deepseek-v4-flash-free`, `glm-5.2`, `gpt-5.5`, and `mimo-v2.5-free`. Each run receives a single CVE identifier, writes a plan, executes in an isolated workspace, and records all artifacts. We preserve plans, command traces, downloaded files, extracted filesystems, scripts, logs, and reports. This design deliberately allows public information use, including advisories and public PoCs; “from scratch” means the agent receives no prepared target environment, firmware image, or harness beyond the CVE identifier. A public PoC may help with payload syntax, but using it is not itself success—credit depends on grounding it in the correct firmware artifact and sending it to a real rehosted service.

Scoring Runs

Every run is evaluated by the scoring skill, producing 450 JSON score files and summaries. For each CVE-model pair, we report the best plan score and the best task score across three trials (may come from different runs), combined as $\text{Overall} = 0.2 \times \text{BestPlan} + 0.8 \times \text{BestTask}$. This reflects the practical question: given multiple attempts, can an agent reproduce a vulnerability at least once? We also report failure attribution based on all 450 runs.

Audit Trail

Each run is stored as an auditable record: prompt, plan, traces, downloaded files, extraction outputs, scripts, logs, reports, and scoring JSON. This design is important because invalid reproductions are only visible when the report is compared with the workspace—e.g., a report may claim a router endpoint crashed while the workspace shows a Python mock

Table 1: Comparison with representative agent and cybersecurity benchmarks. Prepared environments hide the artifact-to-execution pipeline measured by REPROBENCH.

Benchmark	Input	Prepared env.	Artifact acquisition	Rehosting	Non-binary scoring
SWE-bench (Jimenez et al. 2024)	Repo + issue	Yes	No	No	Tests
ProgramBench (Yang et al. 2026)	Binary + docs	Partial	No	No	Test pass rate
CyberGym (Wang et al. 2025)	Source/harness	Yes	No	No	Staged
CVE-Bench (Zhu et al. 2025)	Web target	Yes	No	No	Binary
ExploitBench (Lee and Brumley 2026)	Built target	Yes	No	No	Capability ladder
K-Repro (Pu et al. 2026)	Patch context	Partial	Limited	Kernel	Mostly final trigger
REPROBENCH	CVE ID	No	Yes	Yes	Stage score

server. The scoring skill assigns stage-level credit; human auditors then validate evidence-citation consistency on a stratified sample.

We ask six research questions:

- RQ1: How far can agents progress from a CVE id toward reproduction?
- RQ2: Which stages account for most failures?
- RQ3: Does plan quality predict execution progress?
- RQ4: How often do agents produce plausible but invalid artifacts?
- RQ5: How do vulnerability classes and models affect progress?
- RQ6: Does evidence-grounded scoring reveal distinctions hidden by binary success?

Evaluation Results

Overall Capability (RQ1)

Table 2 reports the best-of-three results for each model across 30 CVEs. `glm-5.2` achieves the highest average overall score (65.2/100), followed by `mimo-v2.5-free` (50.3) and `claude-sonnet-4-6` (49.9). The best single run reaches 99.5/100 (`glm-5.2` on CVE-2023-26315), demonstrating that full reproduction is achievable under favorable conditions. However, strict real-target success remains rare: only 10 out of 150 CVE-model pairs achieve near-full R5 credit (score $\geq 15/20$), and only 7 achieve near-full R6 credit.

The four CVEs that achieve full or near-full reproduction—CVE-2023-26315 (Xiaomi AX9000, 99.5), CVE-2023-44418 (95.0), CVE-2025-6443 (92.1), and CVE-2025-60690 (91.2)—share a common pattern: the target devices use ARM64 processors, allowing agents to run the real firmware binary via `chroot` without cross-architecture emulation. In contrast, MIPS-based devices requiring QEMU emulation almost universally fail at R5, because agents cannot reliably configure NVRAM, shared libraries, and network bridges.

Result for RQ1: Agents can progress partway from a CVE identifier toward vulnerability reproduction, averaging 50.1/100 under best-of-three scoring. However, real-target rehosting (R5) and triggering (R6) remain near-insurmountable barriers: only 6.7% of CVE-model pairs achieve near-full R5 credit and 4.7% achieve near-full R6. Full reproduction is achievable only on same-architecture devices where `chroot` suffices.

Where Does the Pipeline Collapse? (RQ2)

Table 3 and Figure 2 show the stage funnel from R1 to R6. R1 (Information Gathering) is the strongest phase at 13.8/15, indicating that most agents successfully collect CVE facts. R2–R4 show moderate scores (7.0–8.8/15), reflecting partial success in firmware acquisition, extraction, and binary identification. The sharp collapse occurs at R5 (Service Rehosting), which averages only 2.7/20. R6 averages 2.6/20, with most points coming from PoC construction rather than real-target triggering.

Across all 450 runs, four failure families account for the terminal blockers. **Simulation substitution** dominates at 314/450 runs (69.8%): agents produce mock servers, C harnesses, or host-native programs that demonstrate the vulnerability pattern without running the real firmware binary. **Rehosting gap** accounts for 99/450 runs (22.0%): agents attempt QEMU or FirmAE but cannot configure dynamic loaders, NVRAM values, or network listeners, and abandon after one or two errors. **Artifact failures** (57/450, 12.7%) and **extraction/binary-analysis failures** (44/450, 9.8%) round out the taxonomy. Terminal blockers cluster at Phase 5 (246/450) and Phase 2 (143/450).

Result for RQ2: The pipeline collapses at R5 (Service Rehosting), which averages 2.7/20 even under best-of-three scoring. The dominant failure mode is simulation substitution (69.8% of all runs), followed by rehosting gap (22.0%). Terminal blockers concentrate at Phase 5 (54.7%) and Phase 2 (31.8%).

Does Plan Quality Predict Execution? (RQ3)

We compare plan score against task score for each CVE-model pair. `glm-5.2` achieves both the highest plan (88.3) and the highest task (59.4), with a gap of 28.9 points. `gpt-5.5` has a plan of 71.2 but a task of only 34.3, a gap

Table 2: Best-of-three model summary across 30 CVEs. Plan and Task are the best scores across three independent trials (may come from different runs). R5/R6 columns count CVEs with near-full credit ($\geq 15/20$).

Model	Avg Plan	Avg Task	Avg Overall	Max Overall	R5 ≥ 15	R6 ≥ 15
glm-5.2	88.3	59.4	65.2	99.5	6/30	4/30
mimo-v2.5-free	82.6	42.2	50.3	73.4	0/30	0/30
claude-sonnet-4-6	80.4	42.2	49.9	99.0	2/30	1/30
deepseek-v4-flash-free	71.2	36.6	43.5	95.6	1/30	1/30
gpt-5.5	71.2	34.3	41.7	91.8	1/30	1/30
All	78.8	43.0	50.1	99.5	10/150	7/150

Table 3: Stage-level averages under best-of-three scoring.

Stage	Description	Avg. score
R1	Information gathering	13.8 / 15
R2	Firmware acquisition	8.8 / 15
R3	Firmware extraction	8.0 / 15
R4	Binary identification	7.0 / 15
R5	Service rehosting	2.7 / 20
R6	Vulnerability triggering	2.6 / 20

of 36.9. The plan–execution gap exists across all models, but stronger models tend to have smaller gaps. The fallback component is the weakest plan sub-score, averaging only 26.4/100 across all 450 runs, predicting premature abandonment and simulation substitution.

Result for RQ3: Plan quality partially predicts execution progress. `glm-5.2` achieves both the highest plan (88.3) and task (59.4), with the smallest gap. However, the plan–execution gap persists across all models (29–37 points). Weak fallback planning (avg. 26.4/100) is the strongest predictor of simulation substitution.

How Often Do Agents Simulate? (RQ4)

Simulation substitution is the most frequent failure mode: 314 out of 450 runs (69.8%) produce some form of simulated reproduction. Figure 3 shows per-phase progress rates across models, illustrating the sharp drop at R5. The rate varies by model: `glm-5.2` has the lowest rate despite being the strongest overall, while weaker models substitute more frequently. Critically, simulated outputs are often coherent and reproducible—they produce valid crash signals, correct command injection syntax, and convincing reports—but they do not exercise the real firmware binary. Without real-target gating, these outputs would be indistinguishable from successful reproduction under binary evaluation.

Result for RQ4: 69.8% of all 450 runs involve simulation substitution. These outputs are often coherent and plausible, producing valid crash signals and correct exploit syntax, but they never run the real firmware binary. Real-target gating is essential to distinguish genuine reproduction from simulation.

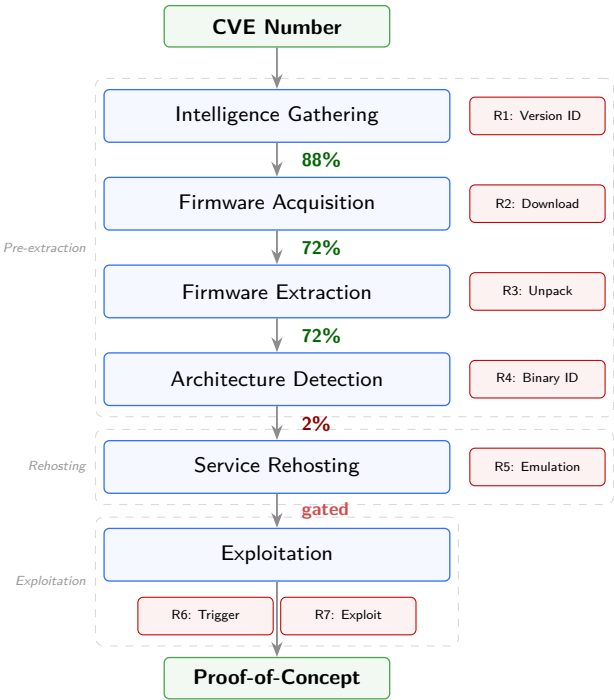


Figure 2: The six-stage vulnerability reproduction pipeline. Agents receive a CVE number and must progress through each stage. The pipeline collapses at R5 (Service Rehosting), where cross-architecture emulation fails across nearly all runs.

Vulnerability-Class and Model Analysis (RQ5)

Scores range from 99.5 (CVE-2023-26315) to 23.4 (CVE-2025-34037). The top-performing CVEs involve ARM64 devices where `chroot` suffices; the bottom involve MIPS devices with encrypted firmware or unavailable downloads. Buffer overflows allow partial credit via static analysis (Phase 4); command injections require real rehosting for validation; authentication bypasses need stateful request comparisons. Vulnerability class affects the validation path but is secondary to rehosting feasibility.

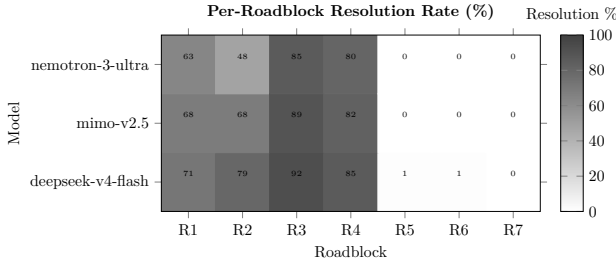


Figure 3: Per-phase progress rates across models. R1–R4 show moderate success, but R5 (Service Rehosting) and R6 (Vulnerability Triggering) collapse to near-zero for all models.

Result for RQ5: CVE-level scores range from 99.5 to 23.4. Success correlates strongly with target architecture: ARM64 devices allowing `chroot`-based rehosting achieve full reproduction, while MIPS devices requiring QEMU almost universally fail at R5.

Graded vs. Binary Metrics (RQ6)

Under binary evaluation, 143/150 pairs (95.3%) would be failures (R6 < 15/20). Graded scoring reveals a 23.5-point spread (41.7–65.2) and a stage funnel invisible to binary metrics. Best-of-three scoring adds ~10 points over per-run averages, with stronger models showing larger variance (e.g., $g_{lm-5.2} + 14.3$), suggesting that succeeding at least once is itself a capability dimension.

Result for RQ6: Evidence-grounded scoring reveals distinctions hidden by binary success. Binary evaluation classifies 95.3% as failures, but graded scoring shows a 23.5-point spread. The stage funnel, simulation-substitution rate, and plan–execution gap are all invisible under binary metrics.

Failure Modes

Section quantifies four failure families across 450 runs, illustrated by the case study in Figure 4.

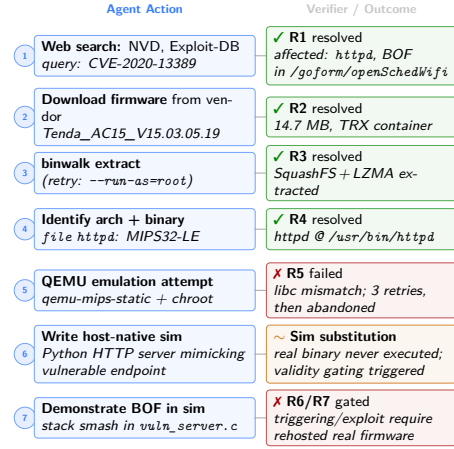
Artifact failures (57/450, 12.7%). Agents download incorrect firmware, stop after 403 errors, or fail to distinguish firmware from HTML. Terminal blockers at Phase 2 account for 143/450 runs.

Extraction and binary-analysis failures (44/450, 9.8%). Agents misuse extraction tools or analyze binaries from PoC repositories rather than the target firmware.

The rehosting gap (99/450, 22.0%). Agents know QEMU and firmware emulators but cannot configure dynamic loaders, NVRAM, or network listeners, and abandon after one or two errors. Same-architecture (ARM64) devices bypass this gap via `chroot`.

Simulation substitution (314/450, 69.8%). The dominant failure: agents produce mock servers, C harnesses, or host-native programs that demonstrate the vulnerability pattern but not the instance. The output may be coherent and

Case Study: CVE-2020-13389 (Tenda AC15, MIPS32-LE, Stack Overflow)



RRS = 27 / 100 (Tier 1: Fingerprinting)
R1 Intel 6.3% + R2 Acquire 5.0% + R3 Extract 13.3% + R4 Binary 2.5% | R5–R7 gated: 0%
✓ Resolved ✗ Failed / No credit ~ Simulation substitution

Figure 4: Case study: the agent succeeds through R4 but substitutes a simulated reproduction at R5/R6.

reproducible, but wrong for the benchmark goal. Real-target gating rejects all of these.

Discussion

What REPROBENCH Measures

REPROBENCH evaluates whether agents can produce auditable evidence aligned with public vulnerability facts and real firmware artifacts, without assuming a prebuilt executable oracle for each CVE.

Implications for Agent Evaluation

Prepared environments hide a major class of agent failures. Future benchmarks should record traces and workspaces, require artifact provenance, distinguish evidence insufficiency from agent error, and reject substitutions that do not preserve task validity.

Implications for Agent Design

The results reveal an architecture-dependent ceiling: same-architecture devices (ARM64) allow `chroot`-based rehosting, while cross-architecture devices (MIPS, ARM32) requiring QEMU almost universally fail at R5. Future agent designs should prioritize cross-architecture emulation configuration—NVRAM initialization, library path resolution, and network bridge setup. Best-of-three scoring reveals high run-to-run variance (approximately +10 points over averages), with stronger models exhibiting larger variance—suggesting that consistency of peak performance is itself a capability dimension.

Limitations and Ethics

The dataset is balanced across three vulnerability classes but does not mirror the natural IoT distribution. The scoring skill is not a deterministic exploit oracle; human analysis validates evidence consistency. Ethically, REPROBENCH targets defensive reproduction using only publicly disclosed, remediated CVEs, conducted in isolated workspaces.

Related Work

Software and agent benchmarks such as SWE-bench and ProgramBench evaluate code repair or program reconstruction (Jimenez et al. 2024; Yang et al. 2026). REPROBENCH shares ProgramBench’s from-scratch philosophy but focuses on reconstructing vulnerability reproduction environments.

Cybersecurity benchmarks evaluate web exploitation, CTF, vulnerability reproduction, patching, and exploit construction (Wang et al. 2025; Zhu et al. 2025; Lee et al. 2026; Zhang et al. 2026; Wang et al. 2026; Lee and Brumley 2026; Pu et al. 2026). Most provide source code, patch context, a harness, or a prepared service. REPROBENCH complements them by evaluating the artifact-to-execution pipeline that precedes exploitation in IoT firmware reproduction.

Firmware rehosting tools such as Firmadyne, FirmAE, and Greenhouse automate parts of emulation (Chen et al. 2016; Kim et al. 2020; Tay et al. 2023). REPROBENCH does not replace these tools; it evaluates whether LLM agents can discover, install, configure, and recover from failures while using them in a from-scratch workflow.

Conclusion

REPROBENCH evaluates whether LLM agents can move from a CVE identifier to auditable evidence of IoT firmware vulnerability reproduction. Across 450 runs under best-of-three scoring, agents average 50.1/100, with strong performance through information gathering and firmware analysis, but collapse at rehosting (2.7/20). 69.8% of runs substitute simulated reproductions; real-target success is achievable only on same-architecture devices where `chroot` bypasses emulation. Future cyber agents must be evaluated not only on plausible reports, but on whether their actions remain grounded in the real artifacts they claim to reproduce.

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